cellular division. If each round requires one day, this type of arithmetic increase would require one million days or 2739.7 years. This arithmetic rate is capable of producing small number of cells present in very small parts of plants. For example the hair on many leaves and stems consists of just a single row of cells produced by the division of the basal cell, the cell at the bottom of the hair next to other epidermal cells. Hair may contain 5 to 10 cells by the division of the basal cell. So, all its cells could be produced in just five to ten days. In the figure 15.2, on plotting the hight of the plant against time a linear curve is obtained. Mathematically it is expressed as:

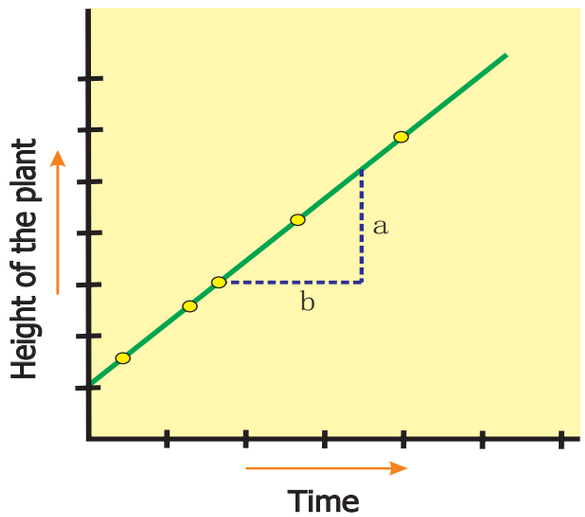



**Figure 15.2:** Constant Linear Growth

$L_t = L_o + rt$

$L_t$ = length at time 't'

$L_o$ = length at time zero

$r$ = growth rate of elongation per unit

**ii. Geometric growth rate:**

This growth occurs in many higher plants and plant organs and is measured in size or weight. In plant growth, geometric cell division results if all cells of an organism or tissue are active mitotically. Example: Round three in the given figure 15.3, produces 8 cells as $2^3 = 8$ and after round 20 there are $2^{20}$ = 1,048,576 cells.

The large plant or animal parts are produced this way. In fact, it is common in animals but rare in plants except when they are young and small. Exponential growth curve can be expressed as,

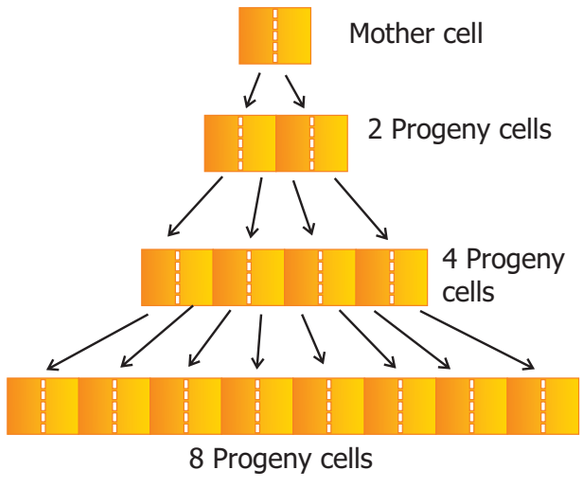



**Figure 15.3:** Geometric growth

$W_1 = W_0 e^{rt}$

$W_1$ = Final size (weight, height and number)

$W_0$ = Initial size at the beginning of the period

$r$ = Growth rate

$t$ = Time of growth

$e$ = Base of the natural logarithms

Here '$r$' is the relative growth rate and also a measure of the ability of the plant to produce new plant material, referred to as efficiency index. Hence, the final size of $W_1$ depends on the initial size $W_0$.

**iii. Arithmetic and Geometric Growth of Embryo**

Plants often grow by a combination of arithmetic and geometric growth patterns. A young embryonic plant grows geometrically and cell division becomes restricted to certain cells at the tips of roots and shoots. After this point, growth is of the slower arithmetic type, but some of the new cells that are produced can develop into their mature condition and begin carrying



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